

Transfer Learning from Historical Monolingual Data for Low-Resource MT: A Case Study of Georgian

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Abstract

This project investigates whether the order of presenting historical monolingual data during continued pretraining of an LLM impacts the performance of machine translation models for low-resource languages, specifically English to Modern Georgian. The four variants of LLaMA 3.2-3B model are: (1) a baseline model fine-tuned for MT from English to Georgian, (2) mixed-order model, pretrained on historical monolingual Georgian data with MT finetuning applied, (3) a chronological model pretrained on ordered historical monolingual data (Old → Middle → Modern) with MT finetuning applied, and (4) a reverse-chronological model pretrained on ordered historical monolingual data (Modern → Middle → Old) with MT finetuning applied. This approach leverages curriculum learning principles to determine the potential it has to enhance translation quality. Results suggest that chronological ordering yields modest improvements in COMET scores and Human evaluation compared to mixed or no historical data, highlighting the value of structured historical data in pretraining for low-resource MT. This work contributes to understanding curriculum-based strategies in continued pretraining for low-resource languages.

1 Introduction

Machine translation (MT) for low-resource languages like Georgian remains challenging due to limited parallel data and monolingual corpora. Georgian, with its unique script and agglutinative morphology, exacerbates these issues, leading to suboptimal performance in neural MT systems. This project is motivated by the potential of historical linguistic data to enrich modern language models, drawing from curriculum learning concepts where data is presented in a structured order to facilitate better learning.

The primary objective is to evaluate if chronological (Old → Middle → Modern) or

reverse-chronological (Modern → Middle → Old) ordering of monolingual Georgian data during continued pretraining improves English to Modern Georgian translation performance more than unstructured mixed ordering or no historical data pretraining at all. The aim is to train and compare four models based on LLaMA 3.2-3B to quantify these effects using standard MT metrics.

The rest of the paper is structured as follows: Section 2 reviews related work, Section 3 describes the datasets used in pre-training and finetuning, Section 4 details the methodology used, Section 5 outlines the work completed, Section 6 presents and discusses the results, Section 7 concludes with future work and offers reflection on spiritual insights garnered through this project.

2 Related Work

Curriculum learning, which involves presenting training data in a structured order, has been explored to enhance the efficiency and performance of language model pretraining. For instance, it has been demonstrated that curriculum strategies accelerate convergence during pretraining phases by optimizing the learning trajectory. [Campos \(2021\)](#) examines linguistically motivated curricula for language modeling, revealing improvements in transfer learning tasks through progressive exposure to a variety of linguistic features.

In the domain of continued pretraining for low-resource machine translation (MT), several approaches focus on adapting large language models (LLMs) to under-resourced languages. [\(Liu et al., 2021\)](#) propose a continual learning framework for multilingual models, enabling adaptation to unseen languages and yielding better fine-tuning outcomes. Building on this, [\(Kumar et al., 2023\)](#) highlight the role of continued pretraining with synthetic data to bolster low-resource

languages, which aligns with the utilization of historical corpora as a natural alternative to synthetic augmentation. A practical application of continued pretraining is seen in (Jasonarson and Steingrímsson, 2025), where lightweight Llama 3.2 models are pretrained on monolingual Icelandic data from sources like OSCAR and the Icelandic Gigaword Corpus before fine-tuning for English→Icelandic MT. Their method employs continual pretraining (CPT) to address vocabulary under-representation in low-resource languages, achieving competitive results despite limited data, though without incorporating historical ordering.

Leveraging monolingual data is a key strategy for mitigating data scarcity in low-resource NMT. Gibadullin et al. (2019) provide a comprehensive survey of methods to exploit monolingual corpora, categorizing them into architecture-independent approaches (e.g., generating pseudo-parallel corpora via back-translation) and architecture-dependent techniques (e.g., integrating language modeling into NMT architectures). Sennrich et al. (2016) emphasize back-translation as an effective method for creating synthetic parallel data from target monolingual sentences, leading to BLEU improvements in low-resource pairs like English→Turkish. Complementing this, data augmentation techniques target rare words to enhance model robustness. Fadaee et al. (2017) introduce translation data augmentation (TDA), which generates new sentence pairs by replacing common words with rare ones using language models, preserving semantic equivalence while diversifying contexts. This approach yields up to 3.2 BLEU gains over back-translation in simulated low-resource settings for bidirectional English↔German.

The use of historical data introduces the opportunity for a diachronic perspective in training, crucial for languages with rich evolutionary histories like Georgian. We employ diachronic word embeddings across historical corpora in four languages to uncover statistical laws of semantic change: the law of conformity (semantic change rates scale inversely with word frequency) and the law of innovation (polysemous words exhibit higher change rates after controlling for frequency). Their analysis, spanning two centuries, underscores how historical monolingual data can

model linguistic evolution, providing a foundation for a curriculum-based ordering to capture such shifts progressively.

This project builds upon these works by integrating curriculum learning into continued pretraining with historical monolingual Georgian data, extending the monolingual leveraging surveyed in Gibadullin et al. (2019) and the CPT adaptation through chronological ordering. Unlike synthetic augmentation in Kumar et al. (2023) or Fadaee et al. (2017), we utilize natural historical texts to address limitations of random or mixed pretraining, such as overlooking evolutionary patterns identified in Hamilton et al. (2018). This approach positions this project within the broader context of low-resource MT, where structured historical data can enhance model adaptation for morphologically rich languages like Georgian.

3 Dataset

For pretraining, the Thesaurus Indogermanischer Text- und Sprachmaterialien (TITUS) corpus, which contains comprehensive corpora for ancient Indo-European languages as well as diachronic language data for certain non-Indo-European languages, including Georgian. This includes sub-corpora for Old Georgian (5th-11th centuries), Middle Georgian (12th-18th centuries), and Modern Georgian (19th century onward). The monolingual data totals approximately several million words across periods, sourced from literary, religious, and contemporary texts.

For parallel English-Georgian data, I used the translation memories of The Church of Jesus Christ of Latter-day Saints, comprising about 100,000 sentence pairs from the religious domain. Preprocessing involved normalization of the Old and Middle Georgian scripts into the Modern Mkhedruli script then tokenization, removal of duplicates, and filtering sentences that are too long. Historical data is segmented by era for ordered pretraining.

4 Methodology

This project’s technical approach consists of two main phases: (1) continued pretraining on monolingual Georgian data, followed by (2) supervised fine-tuning on English→Modern Georgian parallel data for machine translation

(MT). All models are based on Meta’s LLaMA 3.2-3B, a decoder-only transformer architecture with 3 billion parameters. This lightweight LLM serves as the foundation due to its strong generative capabilities, it being open source, and its ability for adaptation, as demonstrated in similar continued pretraining efforts for languages low-resource languages (Jasonarson and Steingrímsson, 2025).

The goal of continued pretraining is to adapt the base LLaMA 3.2-3B model, originally trained primarily on English-dominant data, to the Georgian language, particularly its unique Mkhedruli script and morphological complexity. The standard causal language modeling objective is employed for the pretraining process, where the model predicts the next token in a sequence given all preceding tokens. This is implemented using a left-to-right causal attention mask in the transformer decoder.

Inputs: Long sequences of monolingual Georgian text (concatenated paragraphs or lines from the TITUS corpus, maximum sequence length 2048 tokens).

Outputs: Probability distributions over the vocabulary for the next token; training loss is cross-entropy on ground-truth next tokens.

For pretraining, the following four variants are implemented:

1. Baseline: Direct MT fine-tuning on parallel data without additional pretraining.
2. Mixed: Continued pretraining on shuffled historical and modern data.
3. Chronological: Continued pretraining data presented from Old to Middle to Modern Georgian, epoch-wise.
4. Reverse-Chronological: Continued pretraining data presented from Modern to Middle to Old Georgian, epoch-wise.

Each curriculum stage uses 5 epochs with a learning rate of $1e-5$, global batch size 32 (with gradient accumulation as needed), and AdamW optimizer in bfloat16 precision. Training is performed on an NVIDIA A100 GPU using the Hugging Face Transformers library with Unsloth

optimizations for faster LLaMA training.

The tokenizer is the original LLaMA 3.2 tokenizer, which already provides reasonable coverage of Mkhedruli characters (Unicode block U+10A0–U+10FF); minor vocabulary expansion is applied if rare historical forms are encountered.

After continued pretraining (or directly from the base for the baseline), we perform parameter-efficient fine-tuning for the English→Modern Georgian translation task using Low-Rank Adaptation (LoRA). LoRA adds low-rank update matrices to attention and feed-forward layers (rank 32–64, α 16–32), reducing trainable parameters to approximately 0.5–1% of the full model (15–20M parameters). This makes it easier and faster to train the model.

Task Formulation: Supervised sequence-to-sequence translation framed as autoregressive generation.

Inputs: English source sentence prefixed with a translation instruction prompt (e.g., Translate the following English text to Georgian.)

Outputs: Corresponding Modern Georgian target sentence; cross-entropy loss computed only over target tokens.

Fine-tuning uses Unsloth’s optimized LoRA trainer for training time cut at least in half and resulting in a lessened memory requirement.

Hyperparameters include learning rate $2e-4$ (cosine decay with warmup), effective batch size 64–128, and 5 epochs over the parallel corpus. Early stopping was enabled, though no convergence could be reached.

This two-stage pipeline of domain-adaptive continued pretraining followed by LoRA-based translation fine-tuning, is similar to other low-resource MT adaptations of LLMs while introducing curriculum ordering to explicitly leverage diachronic patterns in Georgian.

5 Experiments

Parallel data is split to have a 5000 line validation set and a 5000 line test set. Monolingual data is segmented by historical era (Old: 5th–11th

Model	BLEU	COMET	MTEval
Baseline	0.05	30.06	3.99
Mixed	0.40	34.54	3.65
Chronological	0.41	36.31	3.24
Reverse	0.86	34.76	3.55

Table 1: MT Performance Metrics. For BLEU and COMET, higher is better. For MTEval, lower is better.

centuries; Middle: 12th–18th; Modern: 19th onward) for curriculum implementation. Additional baselines include an untuned LLaMA 3.2-3B (zero-shot prompted translation) and a standard encoder-decoder Transformer trained from scratch. Evaluation employs SacreBLEU and COMET to capture translation quality.

6 Results & Discussion

The results of this project were far from brimming with hope and clear conclusions to be drawn. Each of the four models did very poorly in translation, however the three pretrained models did have better use of the Georgian script and language. Evaluation results are presented in Table 1.

Chronological ordering outperforms others for COMET and human evaluation, suggesting progressive training aids learning. This aligns with curriculum learning and exceeds mixed data ordering. Limitations include corpus size; unexpected was reverse order’s minor gains. The BLEU score was highest for the Reversed model which is unexpected. This may however indicate why the paper that most chiefly motivated this project, (Takaku et al., 2020) did all pretraining in reverse chronological order. If any preliminary work was done for this paper, and only BLEU was used to determine what order for diachronic word embedding pretraining, perhaps the reverse ordering was chosen based on BLEU scoring.

Had I more compute resources and time, I would have focused on ensuring that the finetuning process for MT were solid and yielded results before moving on to finetune the pre-trained models. Table 2 demonstrates the low quality of translation and Georgian usage. In that example, all three pretrained models were able to output Georgian, however, the output was not of high quality. The baseline did not output the Georgian script, but instead made up Georgian-looking

words using the Latin script. Please note the caption of Table 2 for why all the Georgian characters are not displayed for the non-Baseline models. It should also be noted that there were several examples where the Baseline did output Georgian text that was not complete nonsense. Additionally, there were several instances where the three pretrained models each performed better (giving a higher quality translation) and worse (not using the Georgian script or repeating the English sentence given to it).

Since LLaMA 3.2 does not have any Georgian in its training data, I underestimated how much finetuning and pre-training was required to teach it what to do with this brand new language and brand new script. Because there was a difference seen in the LLM’s usage of Georgian between the three pretrained models and compared to the baseline, I suspect that chronological curriculum training will be superior to the other two methods tested in this paper if the finetuning for translation works properly first.

7 Conclusion & Future Work

Despite the lack of good translational results, we can conclude that pre-training on historical data in general does result in LLM’s being able to output and use the target language in some way.

Future work can further test these hypotheses if more compute is available to ensure an LLM will properly work to translate Georgian.

8 Reflection

This project deepened my understanding of pre-training dynamics and low-resource MT challenges. I gained a better understanding of how LLMs work and the challenges that exist in training and finetuning them especially when it comes to giving it full new data for which it has had no previous context. I have also come to see that importance of our own continual exposure to sources of Truth so that we can more easily receive the revelation God so lovingly gives us. When we receive that revelation, we can do the things He needs us to do in His work of Salvation and Exaltation.

Model	Output	Backtranslation
Source/Target	da gauk'eta sakhls sark'mlebi, rom-lebits igheboda da ik'et'eboda.	And for the house he made windows of narrow lights.
Baseline	Naqvi amsakvelebi saeruli amsakvelebi saeruli amsakvelebi.	-----
Mixed	da misi sakhli iqo sakhli da misi sakhli iqo sakhli da misi.	And his house was a house and his house was a house and his.
Chronological	da mas shina moaktsia samskhverp'lo da misi sakhli da misi.	And he built an altar for him, and his house, and his house.
Reversed	da mistsa mas sakhli da mistsa mas sakhli sakheli da mistsa.	And he gave him a house and gave him the name of the house and gave it to him.

Table 2: The output of the Mixed, Chronological and Reversed models, as well as the target output of the source/target section, were all in Georgian in this example, but LaTeX display issues prevented me from using the Georgian script here while keeping the template the same. These are transliterated version of the Georgian output. There is no backtranslation for the Baseline as it constructed non-Georgian based on Georgian morphemes in the Latin script.

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